

A capacity theorem for horizon entropy: Barrow $\Delta \leq 1$ from a long-range entanglement network

Stilian Pandev 

Independent researcher

E-mail: stilianpandev@gmail.com

Abstract. We model the degrees of freedom of a gravitational horizon as a network of Planck-density sites, each with a bounded total entanglement weight κ (a finite local Hilbert-space dimension) and bond weights decaying with separation as $w(r) \propto r^{-\alpha}$. Three results follow. First, the bond-cut capacity of any region Ω obeys $C(\Omega) \leq \kappa|\Omega|$, so the Barrow entropy deformation satisfies $\Delta \leq 1$ as a theorem of the model (the *bound*; attaining $\Delta = 1$ additionally needs the $\alpha < d$ volume-basin geometry, derived here, and a cut-saturating horizon *state* — the one assumption the theorem does not remove) — no fractal-horizon geometry is needed, and for long-range couplings ($\alpha < d$) none is available, since the minimum cut is not a surface. Second, the capacity is the enclosed *volume* ($\Delta = 1$) for every range $\alpha < d$ and the boundary *area* ($\Delta = 0$) for $\alpha > d + 1$; gravity’s Newtonian reach ($\alpha = 1$) lies deep in the volume phase, and no gravitational coupling produces an intermediate range, so a measured intermediate Δ falsifies the picture rather than selecting a parameter. Third, $\Delta = 1$ exactly is the infrared-cutoff endpoint $L \rightarrow \infty$ — a stated choice, not a derivation. What the network counts is a structural capacity (not manifestly the generalized entropy), and the volume identification is the non-default, empirically-bet-on branch: DESI DR3 + Euclid decide $\Delta = 1$ versus 0 at a forecast $\sigma(\Delta) \approx 0.09$. This shortens — it does not eliminate — the assumption list behind the volume-law postulate of Structural Entropy Dark Energy.

Keywords: holographic dark energy; horizon thermodynamics; entanglement entropy; de Sitter holography; Barrow entropy; long-range systems; apparent horizon; dark energy

1. Introduction

Structural Entropy Dark Energy (SEDE) reproduces the cosmological data with no fitted dark-sector parameter, at the cost of a single foundational input: the cosmic apparent horizon carries *volume-law* entropy, $S \propto V$ (Barrow deformation $\Delta = 1$), rather than the Bekenstein–Hawking area law. The cosmology paper [1] fixes the background from this input and confronts it with the data; the foundations paper [2] asks how much of the input is irreducible, decomposing it into *state* (the horizon degrees of freedom are maximally entangled), *form* (a thermalised state has volume-law entanglement), *scale* (the Cohen–Kaplan–Nelson bound fixes $\rho_{\text{DE}} \sim \rho_{\text{crit}}$), and *count* (whether the entropy counts the enclosed *bulk* degrees of freedom or the *boundary* ones). The first three reduce to established physics; the count — bulk versus boundary — is the residue the foundations paper names as irreducible, a de Sitter holography question with no shortcut past the observer-algebra trace calibration.

This paper reduces the count to a shorter list of stated assumptions; it does not fully discharge it. We model the horizon degrees of freedom as a network with a finite per-site *leg budget* κ — equivalently, a finite local Hilbert-space dimension — and bond weights that decay as a power, $w(r) \propto r^{-\alpha}$. From this single, weaker premise three results follow (Figure 1), each machine-checked (§Reproducibility):

1. $\Delta \leq 1$ is a theorem (§2.2). The bond-cut capacity of any region obeys $C(\Omega) \leq \kappa \cdot |\Omega|$ identically — the entropy a region can carry cannot exceed its degree-of-freedom count. No fractal ceiling and no premise that the horizon is a space-filling surface are needed; $\Delta > 1$ is impossible, and a high-significance $\Delta > 1$ would falsify the framework with no parameter to absorb it.
2. The count is binary, and gravity is on the volume side (§2.3–§2.4). The capacity exponent is the enclosed *volume* ($\Delta = 1$) for every coupling of range $\alpha < d$ and the *area* ($\Delta = 0$) for $\alpha > d + 1$, with the crossover at $\alpha = d$ exactly — the same threshold that separates additive from non-additive long-range statistical mechanics and at which the entanglement area-law theorems [3] lose their hypothesis (§3). The α -controlled volume→(fractal)→area crossover itself is established in the long-range-entanglement literature [4–6] and, for the network cut-scaling itself, in long-range percolation and small-world graph theory [7–9]; our contribution is to read a Barrow index $\Delta = q - 2$ off it for a *horizon* count, not the crossover. Gravity’s $1/r$ reach sits at $\alpha = 1$, deep in the volume basin (§4); the equilibrium, area-law frameworks are its local limit. Whatever selects $\Delta = 1$ must therefore be a *saturation* argument.
3. $\Delta = 1$ as the infinite-cutoff endpoint (§5). The only source of a deficit below the volume law is a finite infrared cutoff L on the network; nothing *internal* to SEDE fixes L , so Δ is a monotone function of it, and $\Delta = 1$ is the $L \rightarrow \infty$ endpoint suggested by the flat, infinite spatial slice. This is a *choice of cutoff* — the identification of the cosmological spatial slice with the entanglement-network infrared cutoff — not a derivation (§5.3–§5.4); $\Delta = 1$ remains the theory’s one postulate, restated at the endpoint.

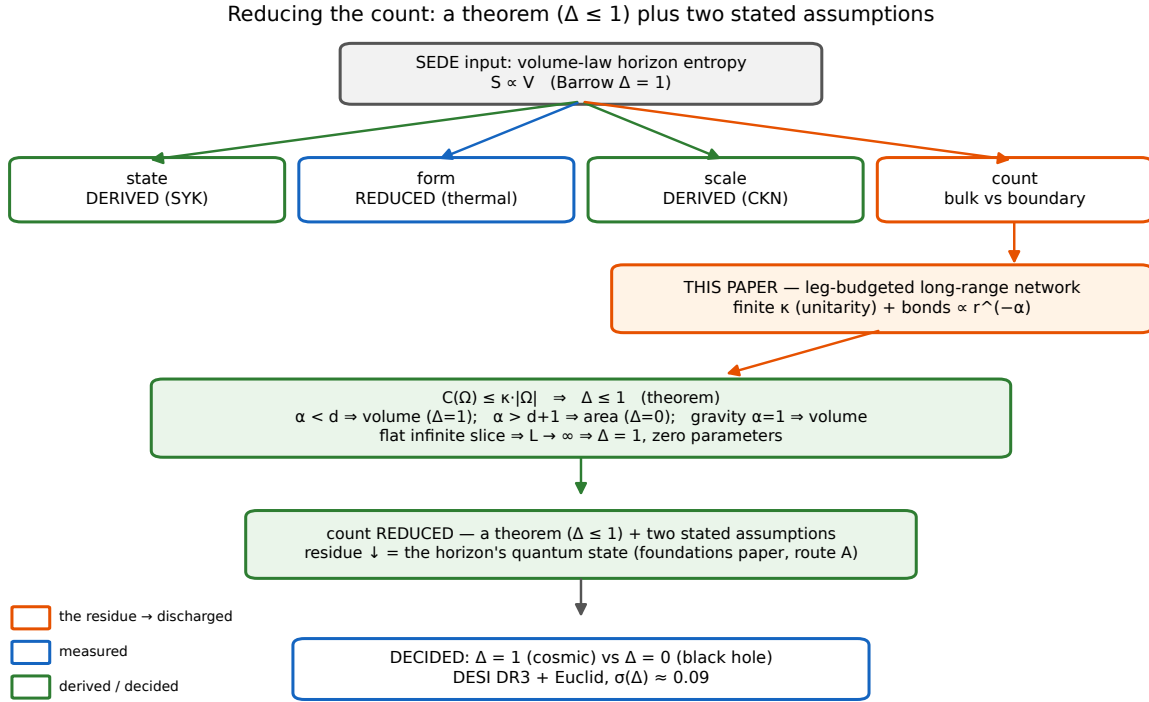


Figure 1. The reduction. SEDE’s one input — volume-law horizon entropy — decomposes into *state*, *form*, *scale*, and *count* (the foundations paper); the first three reduce to established physics, and the *count* is the irreducible residue. Modelling the horizon degrees of freedom as a leg-budgeted long-range network reduces it: $C(\Omega) \leq \kappa \cdot |\Omega|$ makes $\Delta \leq 1$ a theorem (§2.2), $\alpha < d$ puts gravity on the volume branch (§2.3–§4), and $\Delta = 1$ is the $L \rightarrow \infty$ endpoint of the flat infinite slice — a cutoff choice, not a derivation (§5). What survives is the horizon’s quantum *state* (route A of the foundations paper) and that endpoint identification, decided — with the black-hole $\Delta = 0$ — by DESI DR3 + Euclid.

The residue that survives is *strictly weaker* than the postulate it replaces (§7): a finite leg budget (unitarity), a bond weight decaying slower than r^{-d} (an inequality, not a functional form), the infinite-cutoff identification $L \rightarrow \infty$ (a stated choice, §5.3), and the maximally-scrambled state of route A of the foundations paper (untouched). What is bought is that the open content of SEDE now sits in two named places — the horizon’s quantum *state* and the $L \rightarrow \infty$ endpoint reading of its *count* — and, for the one horizon we have, is decided by a scheduled measurement: $\Delta = 1$ for the driven cosmic horizon and $\Delta = 0$ for equilibrium (black-hole) horizons, separated by DESI DR3 + Euclid at a forecast $\sigma(\Delta) \approx 0.09$. We are explicit about what we do not claim: the network is a *model* of the horizon degrees of freedom, not a derivation of them from a fundamental theory, and the bond cut is a *capacity*, an upper bound whose attainment is the state question (§6). What the paper delivers is the counting residue reduced to a theorem (the ceiling) plus two honestly-stated assumptions (the network model with its range inequality, and the $L \rightarrow \infty$ endpoint choice), and a geometry — the count is not a Hausdorff dimension — sharp enough to be falsified.

2. The count from a leg-budgeted long-range network

2.1. Setup

Let the horizon degrees of freedom be sites of a network filling a ball of radius L at number density $n = \ell_P^{-d}$, $d = 3$, with symmetric bond weights

$$w_{ij} \propto r_{ij}^{-\alpha}, \quad \sum_j w_{ij} = \kappa \quad \text{for every } i. \quad (2.1)$$

The budget in (2.1) is not an idealisation. In a tensor-network representation the leg weight attached to a site is $\ln(\text{local Hilbert-space dimension})$; unitarity bounds it. Given the shape $r^{-\alpha}$, the budget is imposed by the unique symmetric scaling $w = \text{diag}(d) W \text{diag}(d)$ with prescribed row sums — the symmetric Sinkhorn problem, which for positive symmetric W has a unique positive solution. All numbers below are computed at Sinkhorn residual $< 10^{-9}$.

We take as the count the bond-cut capacity of a region Ω ,

$$C(\Omega) = \sum_{i \in \Omega, j \notin \Omega} w_{ij}, \quad (2.2)$$

the standard upper bound on the entanglement entropy of Ω with its complement. C is a capacity, not an entropy: whether it is attained depends on the state (§6). Everything in this section concerns C .

2.2. The capacity theorem

From (2.1) and (2.2), for any region Ω ,

$$C(\Omega) \leq \sum_{i \in \Omega} \sum_j w_{ij} = \kappa |\Omega|. \quad (2.3)$$

The entropy of a region cannot exceed its degree-of-freedom count. Writing $S \propto R^{2+\Delta}$ in the Barrow parametrisation $S = (A/A_0)^{1+\Delta/2}$, (2.3) gives

$$\Delta \leq 1. \quad (2.4)$$

This is the ceiling of the cosmology paper (§8.1) [1] — obtained here without reference to the Hausdorff dimension of the horizon, and without the premise that the horizon is a fractal 2-surface. Finite local dimension is the whole content. The theorem is model-conditional: it is a statement about the leg-budgeted network of §2.1 (and, for the $\alpha = d$ crossover of §2.3, one limit interchange remains unproven — §8), not about an arbitrary horizon. Two consequences are worth stating plainly. First, $\Delta > 1$ is impossible in this framework, so a high-significance measurement of $\Delta > 1$ falsifies it with no

parameter available to absorb the result. Second, whatever selects $\Delta = 1$ must be a *saturation* argument, since (2.3) is the only bound in play.

2.3. The crossover is at $\alpha = d$

Consider a site at radius s inside a sphere of radius $\rho \ll L$. The bond mass it places at separations in $[u, u+du]$ scales as $u^{d-1-\alpha} du$. Its outward fraction — the share of its budget spent on bonds crossing ρ — is therefore governed by

$$\int^L u^{d-1-\alpha} du \quad \text{versus} \quad \int^\rho u^{d-1-\alpha} du. \quad (2.5)$$

For $\alpha < d$ the integral is dominated by its upper limit: the bond mass sits at the largest separations, the outward fraction tends to unity for every interior site, and

$$C(\rho) \rightarrow \kappa n V(\rho) \quad (\text{volume law, budget saturated}). \quad (2.6)$$

For $\alpha > d$ the integral converges at small u : only a surface layer of thickness $O(\ell_p)$ contributes, and $C(\rho) \propto A(\rho)$ (area law). The crossover is at $\alpha = d$ exactly — the same threshold that separates additive from non-additive long-range systems (Campa–Dauxois–Ruffo [10]), and the same one at which the hypothesis of the entanglement area-law theorems (Hastings [11]; Brandão–Horodecki [12]) fails.

Numerical confirmation (Figure 2). Spherical reduction with the exact angular average of $|x - y|^{-\alpha}$, ambient $L = 200$, $\kappa = 1$. The table gives $s(\rho) \equiv C(\rho)/N_{\text{in}}$, the capacity per enclosed degree of freedom, and q from $C \propto \rho^q$:

Table 1: Capacity per enclosed degree of freedom $s(\rho) = C(\rho)/N_{\text{in}}$ and measured capacity exponent q ($C \propto \rho^q$) versus bond range α , in the spherical reduction ($L = 200$, $\kappa = 1$).

α	$\rho = 2$	$\rho = 5$	$\rho = 10$	$\rho = 20$	$\rho = 40$	q
1.0	0.9999	0.9996	0.9983	0.9933	0.9728	2.99
2.0	0.9969	0.9873	0.9707	0.9371	0.8690	2.96
2.5	0.9805	0.9375	0.8850	0.8089	0.6994	2.89
4.0	0.7088	0.4156	0.2569	0.1518	0.0863	2.30
6.0	0.5135	0.2206	0.1119	0.0562	0.0281	2.03

At $\alpha = 6$ the capacity per site halves exactly when ρ doubles: $s \propto 1/\rho$, the area law, to three figures. At $\alpha = 1$ it sits at κ to four decimals: every degree of freedom spends essentially its entire budget outward. The volume law is not imposed by a ceiling; it is attained.

The $\alpha = 4$ row warrants a word, because its $q = 2.30$ has not reached 2 and could be mistaken for a failure to converge. It is not: $\alpha = 4 = d + 1$ is the *marginal* area point,

where the capacity-carrying surface layer has a logarithmic tail, $C(\rho) \propto \rho^2 \cdot \ln \rho$, so the finite- ρ exponent is $q(\rho) = 2 + O(1/\ln \rho)$ — inflated above 2 but *decreasing*, not a stable power. Pushing the ambient size out ($L = 1500$, ρ up to 180) settles it with a normalisation-free diagnostic, $C(\rho)/\rho^2$: at $\alpha = 4$ it rises linearly in $\ln \rho$ (*slope* $\approx 2.9 > 0$), the signature of the marginal logarithm, whereas at $\alpha = 5$ — just past the marginal point — it is flat to 6 % with $q = 2.01$, the clean area law. No α produces a *stable* exponent between 2 and 3: the $\alpha = 4$ elevation is a marginal logarithm on the area side, and it sharpens rather than softens the binary dichotomy of §2.4. (For $d < \alpha < d + 1$ the same analysis gives a genuine but still non-volume $q \rightarrow 3 - (\alpha - d)$; at $\alpha = 3.5$, q climbs toward 2.5. The volume phase, $q = 3$, is reached only for $\alpha < d$.)

Identifying the saturated capacity with a constant entropy density,

$$s_0 = \kappa n = \kappa / \ell_P^3, \quad (2.7)$$

recovers eq. (2.2) of the cosmology paper [1], $s_{\text{grav}} = s_0$, with the constant identified as legs per Planck cell.

2.4. The dichotomy is binary — for a gravitational horizon

Equation (2.6) holds for every $\alpha < d$, not merely $\alpha = 1$: the volume phase is a basin, not a knife-edge. The model therefore never requires the specific form $w(r) = 1/r$. It requires only

$$w(r) \text{ decays slower than } r^{-d}. \quad (2.8)$$

Any coupling of gravitational reach satisfies (2.8) with room to spare. We are careful about what is and is not binary. The capacity exponent is $q = 3$ (volume) for *every* $\alpha < d$ and $q = 2$ (area) for $\alpha > d + 1$; the intermediate window $d < \alpha < d + 1$ does interpolate, $q = 3 - (\alpha - d)$, with $\alpha = d + 1$ the marginal area case ($q \rightarrow 2$ with a logarithm) — the crossover is a genuine continuum in α (§2.3), not a two-valued function. What is binary is the count for a *gravitational* horizon: gravity is $1/r$, so $\alpha = 1 \ll d$, sitting deep in the volume basin, while the equilibrium/area frameworks are local (effectively $\alpha > d + 1$). An intermediate Δ would require an intermediate-*range* horizon coupling, $d < \alpha < d + 1$, for which there is no physical mechanism — gravity supplies no such exponent. A measured intermediate Δ therefore falsifies the gravitational-network picture — no coupling of gravitational reach produces it — rather than selecting a tunable exponent. In particular it closes the door on the Kardar–Parisi–Zhang branch ($\Delta \approx 0.61$) once floated as a distinguishable alternative in the foundations paper: §2.5 shows there is no roughening universality class to select in the first place.

2.5. For $\alpha < d$ there is no minimum-cut surface

The Ryu–Takayanagi prescription reads entanglement entropy off a *minimal cut surface*. That object exists only when the bonds are local. We test this directly by maximum flow

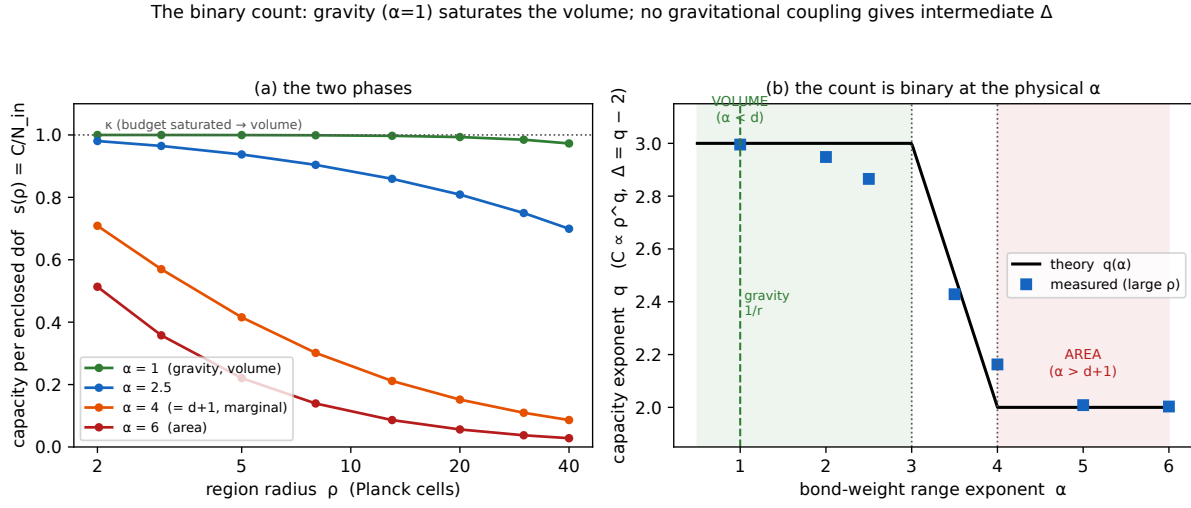


Figure 2. The binary count (§2.3–2.4). (a) Capacity per enclosed degree of freedom $s(\rho) = C/N_{\text{in}}$: at $\alpha = 1$ (gravity) it saturates the leg budget κ (the volume law); at $\alpha = 6$ it falls as $1/\rho$ (the area law). (b) The capacity exponent q (with $\Delta = q - 2$) versus the bond-range α : $q = 3$ (volume) for every $\alpha < d$ and $q = 2$ (area) for $\alpha > d + 1$, interpolating only in the *unphysical* window $d < \alpha < d + 1$ (the marginal $\alpha = d + 1 = 4$ point carries a logarithm, so its finite- ρ value sits just above 2). Gravity’s $1/r$ reach ($\alpha = 1$) lies deep in the volume basin; no gravitational coupling yields an intermediate Δ .

on a lattice ball, with boundary legs of the upper cap tied to a source and the lower cap to a sink at infinite capacity, so that every cut passes through the bulk. Two diagnostics, neither of which depends on any normalisation:

(i) Thickness of the severed-bond set, in lattice units (weighted RMS of the midpoint height of cut bonds):

Table 2: Thickness of the severed-bond set (weighted RMS midpoint height, lattice units) versus ball radius R for each bond range α , from the maximum-flow minimum cut on the lattice ball.

α	R=3	R=4	R=5	R=6
0.5	0.78	1.20	1.60	1.98
1.0	0.82	1.27	1.70	2.11
2.0	0.86	1.43	1.92	2.39
3.0	0.57	1.54	2.11	2.68
4.0	0.55	0.59	0.63	0.64
6.0	0.516	0.516	0.520	0.518

For $\alpha > d$ the thickness is R -independent — a sharp interface one lattice spacing thick, and the cut is the equatorial disk ($|S|/N \rightarrow 0.447$, a half-ball). For $\alpha \leq d$ it grows

No minimum-cut surface for a long-range horizon $\Rightarrow$ no height field, no roughening (§2.5)

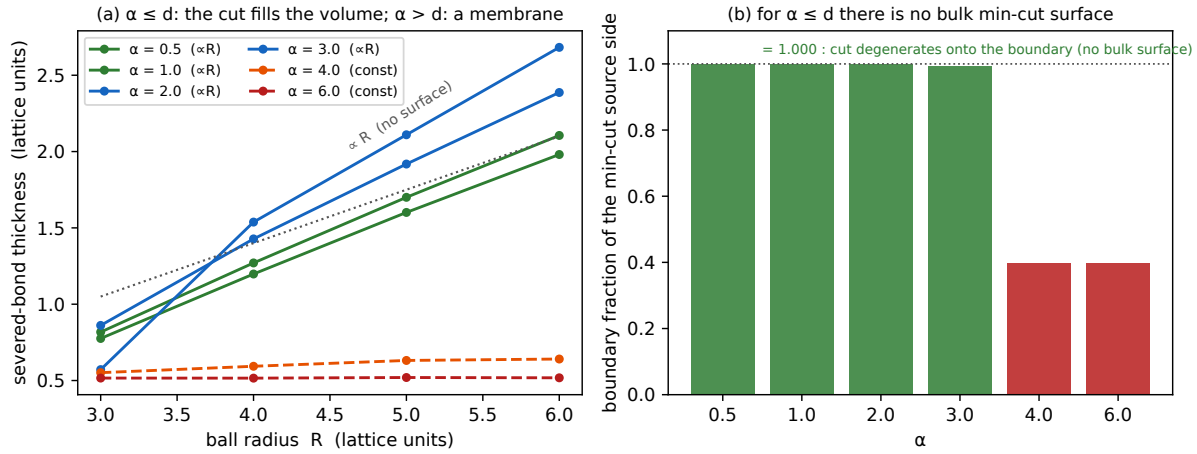


Figure 3. No minimum-cut surface for a long-range horizon (§2.5). (a) The thickness of the severed-bond set grows $\propto R$ for $\alpha \leq d$ (the cut fills the volume — there is no surface) and is R -independent for $\alpha > d$ (a one-cell membrane). (b) The boundary fraction of the minimum cut's source side is 1.000 for $\alpha \leq d$ (the cut degenerates onto the boundary shell) and ≈ 0.40 for $\alpha > d$ (a genuine bulk membrane). With no min-cut surface there is no height field, no roughness exponent, and no roughening — so Δ is a count, not a Hausdorff dimension.

$\propto R$: the severed bonds fill the volume (Figure 3).

(ii) Where the cut lives. For every $\alpha \leq 3$ and every R tested, the fraction of the source side of the minimum cut consisting of boundary sites is 1.000, exactly: the cut degenerates onto the boundary shell. There is no bulk surface, because there is no emergent bulk geometry. (The cut cost then scales as $R^{5-\alpha}$ — super-volume — which is the signature of the *unbudgeted* network's illegitimacy: per-site leg weight $\sum_j r_{ij}^{-1} \sim R^2$ diverges, and no finite local Hilbert space supports it. Imposing (2.1) removes the pathology and returns (2.6).)

The consequence is structural. With no minimum-cut surface there is no height field, no roughness exponent, and no roughening universality class. Δ is a count, not a Hausdorff dimension, exactly as the cosmology paper (§8.5) maintains. This is also the foundations paper's position: it reads the count as the bare bound $\Delta \leq 1$, which §2.2 here *derives* — so the fractal reading $\Delta = d_H - 2$ is unnecessary.‡

An independent confirmation from the classical side: in linearised GR the apparent-horizon condition $\theta_+ = 0$ is, in Newtonian gauge (where the constant-time slicing is shear-free, so K_{ij} is pure trace), the constant-mean-curvature condition $\mathcal{M} = 2H$. It is the Euler-Lagrange equation of $A[h] - 2H \cdot V[h]$, and since the area functional $A =$

‡ The ϵ -lag / roughening observable of the companion papers is superseded here: eq. (2.9) leaves the classical apparent horizon with no growth equation to roughen with. The companion papers already reflect this — the foundations paper reduces its membrane section to the exact coefficients and withdraws the roughening layer, and the cosmology paper withdraws the corresponding predictions (its §6).

$\bar{R}^2 \int d\Omega [(1+h)^2 + \frac{1}{2} |\nabla_S h|^2 + O(|\nabla_S h|^4)]$ carries no cubic term, the constraint reads

$$\nabla_S^2 h + 2h + 2h^2 = 2\mathcal{S}, \quad \mathcal{S} = \Phi + \Psi + \dot{\Psi}/H - H^{-1} \partial_R \Psi. \quad (2.9)$$

There is no $(\nabla h)^2$ term at second order, and — decisively — no $\partial_t h$ at any order. The operator $\nabla_S^2 + 2$ is elliptic on the sphere; $h_{\ell m} = 2\mathcal{S}_{\ell m}/(2 - \ell(\ell + 1))$ is slaved instantaneously to the matter perturbations, and high multipoles are suppressed as $1/\ell^2$. The classical apparent horizon does not roughen. It has no growth equation to roughen with. That the apparent-horizon shape can be found this way — perturbatively, order by order, as a slaved constraint within the canonical formalism — is independently established by Neuser and Thiemann, who compute it to second order in gauge-invariant perturbation theory [13].

This operator is not ours to invent: $\nabla_S^2 + 2$ is the Andersson–Mars–Simon marginally-outer-trapped-surface (MOTS) stability operator [14] specialised to the FRW apparent horizon. On the round 2-sphere (areal radius $\bar{R} = 1/H$) the AMS operator $L\psi = -\nabla_S^2 \psi + 2 s^A D_A \psi + (\frac{1}{2} R_S - |\sigma^+|^2 - s_A s^A + D_A s^A - 8\pi T_{ab} \ell^a k^b) \psi$ collapses — spherical symmetry gives $s_A = 0$, the horizon is shear-free ($\sigma^+ = 0$), and for the within-slice shape variation it reduces to the classical mean-curvature Jacobi operator $-(\nabla_S^2 + |A|^2)$ with $|A|^2 = 2/\bar{R}^2$, i.e. exactly $-(1/R)(\nabla_S^2 + 2)$. Its spectrum is the published one, $\ell(\ell + 1) - 2$, and the $\ell = 1$ kernel is the translation zero mode every MOTS stability operator carries — not a coincidence but the sphere’s freedom to be displaced. We reproduce this independently, by a second geometric route (surface-of-revolution principal curvatures) matched to the spectrum, so the GR leg of this argument rests on a published 2008 theorem rather than on our own symbolic algebra (`ams_stability.py`).

Slicing caveat. The clean *elliptic* (∂_t -free) form is specific to the shear-free Newtonian slice with flat spatial sections. The AMS matter term for FRW is $8\pi T_{ab} \ell^a k^b = 3H^2 + \dot{H}$, so the *null-direction* (marginally-outer-trapped-tube, i.e. time-evolution) variation carries an $\dot{H}$ that the within-slice shape operator does not: the two differ by exactly the horizon’s evolution term. It is the within-slice shape equation — the one governing whether the horizon *roughens* — that is elliptic and slaved; the ∂_t enters only the orthogonal question of how the horizon moves between slices.

3. The premise: finite local dimension, and why $\alpha < d$ evades the area law

The two ingredients of §2 — a finite leg budget κ and a bond weight decaying slower than r^{-d} — are not exotic; each is weaker than something already standard.

Finite κ is unitarity. In a tensor-network representation of the horizon state, the weight on a leg attached to a site is the logarithm of the local bond dimension, $\ln \chi$, which a finite-dimensional local Hilbert space bounds. The per-site budget $\sum_j w_{ij} = \kappa$ of (2.1) is therefore the statement that each horizon degree of freedom has a bounded amount of entanglement to distribute — a consequence of unitarity, not an assumption

about geometry. The bond cut $C(\Omega)$ is then the standard Ryu–Takayanagi / minimum-cut upper bound on the entanglement entropy of Ω (§6.1). Everything in §2.2–§2.4 uses only this budget and the *range* of the weights; it never uses the specific form $1/r$.

Why $\alpha < d$ is allowed — the area-law theorems and their hypothesis. A natural objection is that entanglement entropy “always” obeys an area law, so a volume-law horizon is forbidden. It is not: the entanglement area-law *theorems* are theorems about *local* systems. In one dimension, gapped local Hamiltonians have area-law (there, constant) ground-state entanglement [11]; the higher-dimensional results [12] likewise assume short-range interactions and a spectral gap. Their load-bearing hypothesis is exactly locality — couplings that fall off fast enough that a region’s boundary dominates its entanglement — and that hypothesis fails precisely when the weights decay slower than r^{-d} : the same $\alpha = d$ threshold at which our capacity crossover sits (§2.3) and at which long-range statistical mechanics stops being additive [10]. A volume-law entanglement entropy is thus not a violation of anything; it is the generic expectation once the connectivity is long-range. The premise (2.8) is the *negation* of the area-law theorems’ hypothesis, and the horizon — connected by gravity — is on the far side of it.

The reduction is therefore in the *strength* of what is assumed. The postulate replaced was “the horizon entropy counts its volume.” What remains is “the horizon degrees of freedom have finite local dimension and a longer-than-marginal-range coupling,” plus the state assumption of §6.1. The first is unitarity; the second is an inequality gravity satisfies with two decades of margin (§4); only the third is a genuine physical input, and it is the foundations paper’s, not ours.

4. Gravity lands in the volume phase

Which side of the $\alpha = d$ crossover is the cosmic horizon on? The bond weights of (2.1) encode which horizon degrees of freedom are entangled with which, and how strongly, as a function of separation. For a horizon whose microscopic correlations are set by gravity, that reach is Newtonian, $w(r) \propto 1/r$, so $\alpha = 1$ — well below $d = 3$, deep in the volume basin (2.6), with a margin of two full units in the exponent. This is why the dichotomy of §2.4 is not delicate for the physical case: gravity does not sit near the crossover, and no $O(1)$ uncertainty in the effective coupling could carry it across two decades to the area side.

That same long-rangedness is the object the foundations paper works with from the thermodynamic direction. There the horizon degrees of freedom carry a cooperative coupling $J = \lambda_{\max}(W_{\text{grav}})$ — the largest eigenvalue of the very weight matrix $W_{ij} \propto r^{-\alpha}$ of (2.1) — and $\alpha < d$ is exactly the regime in which that coupling is *super-extensive* ($\lambda_{\max} \propto N^{1-\alpha/d}$) and mean-field thermodynamics is exact [10]. So the network’s “ $\alpha < d \Rightarrow$ volume” of the present paper (the *count*) and the foundations paper’s “non-additive cooperativity $\Rightarrow$ a bistable area $\leftrightarrow$ volume free energy $F(m)$, selected at the horizon’s birth” (the *selection*) are two readings of one fact: gravity is long-range at the horizon. The count fixes that the two branches are exactly {area, volume} and that $\Delta \leq 1$; the selection fixes

which branch a given horizon occupies — the grown cosmic horizon carried through the ordering bifurcation onto the volume branch (a *tilted* pitchfork, whose absence of branch ambiguity is the generic feature of its universal unfolding [15]), the abruptly-formed black hole locked on the area branch. The two papers meet here: this one fixes the ceiling and the two available branches of the count, the foundations paper the dynamics that selects between them.

5. The edge correction and the infrared cutoff

5.1. The exact deficit law

Equation (2.6) is exact only for $\rho \ll L$. At finite ρ/L the outward fraction falls short, because the bond mass a site can place beyond ρ is limited by the ambient size. From (2.5), for $\alpha < d$,

$$1 - s(\rho)/\kappa = D(x) = c x^{d-\alpha}, \quad x \equiv \rho/L. \quad (5.1)$$

At $\alpha = 1$ no short-distance regulator is needed: Newton's theorem gives the exact angular average $\langle |x - y|^{-1} \rangle = 1/\max(s, s')$, and $D(x)$ is a pure number. Solving the Sinkhorn problem in the radial reduction and refining the discretisation:

Table 3: Convergence of the deficit coefficient $D(x)/x^2$ with the radial discretisation M , at $\alpha = 1$ (exact Newton-kernel angular average).

M	$D(0.02)/x^2$	$D(0.03)/x^2$
750	0.6738	0.6439
1500	0.6732	0.6732
3000	0.67308	0.67316
6000	0.67305	0.67315

$$c = \lim_{x \rightarrow 0} \frac{D(x)}{x^2} = 0.6730 \pm 0.0001,$$

with a fitted exponent 2.0018 (predicted $d - \alpha = 2$). (Note that D/x^2 drifts upward with x — 0.6731 at $x = 0.02$, 0.6817 at $x = 0.20$ — so c must be read as the small- x limit, not as the intercept of a power-law fit over a finite window.)

The exponent $d - \alpha$ is confirmed independently at $\alpha = 2$ (*deficit* $\propto x$) and $\alpha = 2.5$ ($\propto x^{1/2}$). We record that c is *not* $2/3 = 0.66667$, from which it differs by 1%, well outside the discretisation error. We know of no closed form and do not assert one.

5.2. The effective deformation

With $S(\rho) = \kappa \mathbf{n} V(\rho)[1 - D(\rho/L)]$,

$$\Delta(x) = \frac{d \ln S}{d \ln \rho} - 2 = 1 - \frac{x D'(x)}{1 - D(x)} = 1 - \frac{(d - \alpha) c x^{d-\alpha}}{1 - c x^{d-\alpha}}. \quad (5.2)$$

At $\alpha = 1$ (Figure 4):

Table 4: Effective deformation Δ versus the ratio of horizon radius to network infrared cutoff, R_H/L , at $\alpha = 1$ (eq. 5.2).

R_H/L	Δ	
0.001	1.0000	network far exceeds the horizon
0.10	0.9864	$L = 10 R_H$
0.3125	0.8546	$L = \text{comoving particle horizon}$
0.50	0.5522	$L = 2 R_H$

The deficit is one-signed: $D \geq 0$ by (2.3), so the correction always pushes Δ *downward*, toward the area law, and never above the ceiling. This is consistent with, and is the mechanism behind, (2.4).

5.3. L is not fixed by the model

Expression (5.2) is a one-parameter family in L . Nothing in SEDE — not the CKN bound, not flatness, not the structure gate, not the birth-selection chain of the foundations paper’s §5 — determines the infrared cutoff of the horizon’s degree-of-freedom network. Adopting any finite L therefore introduces a continuously-sampled dark-sector parameter, which is precisely what the model’s central claim forbids (the cosmology paper’s §3: *the dark sector adds no fitted parameter*).

The only value of L consistent with the model’s own background is $L \rightarrow \infty$. The spatial slice of flat FRW is infinite; its flatness is already used, in the cosmology paper’s §2, to normalise $\Omega_{DE}0$. With $L \rightarrow \infty$, $D \rightarrow 0$ and

$$\Delta = 1 \quad \text{exactly.} \quad (5.3)$$

We are explicit about the status of this step, because it is where the value of Δ actually enters. The argument above eliminates the would-be parameter; it does not derive its value. The step that fixes L is the identification of the cosmological spatial slice with the infrared cutoff of the entanglement network — an identification nothing in SEDE determines. $\Delta = 1$ is therefore the *endpoint reading*: a cutoff choice made for consistency with the model’s own no-fitted-parameter claim, under which the pre-registered point carries no free parameter — and it remains the theory’s one postulate, restated, not a

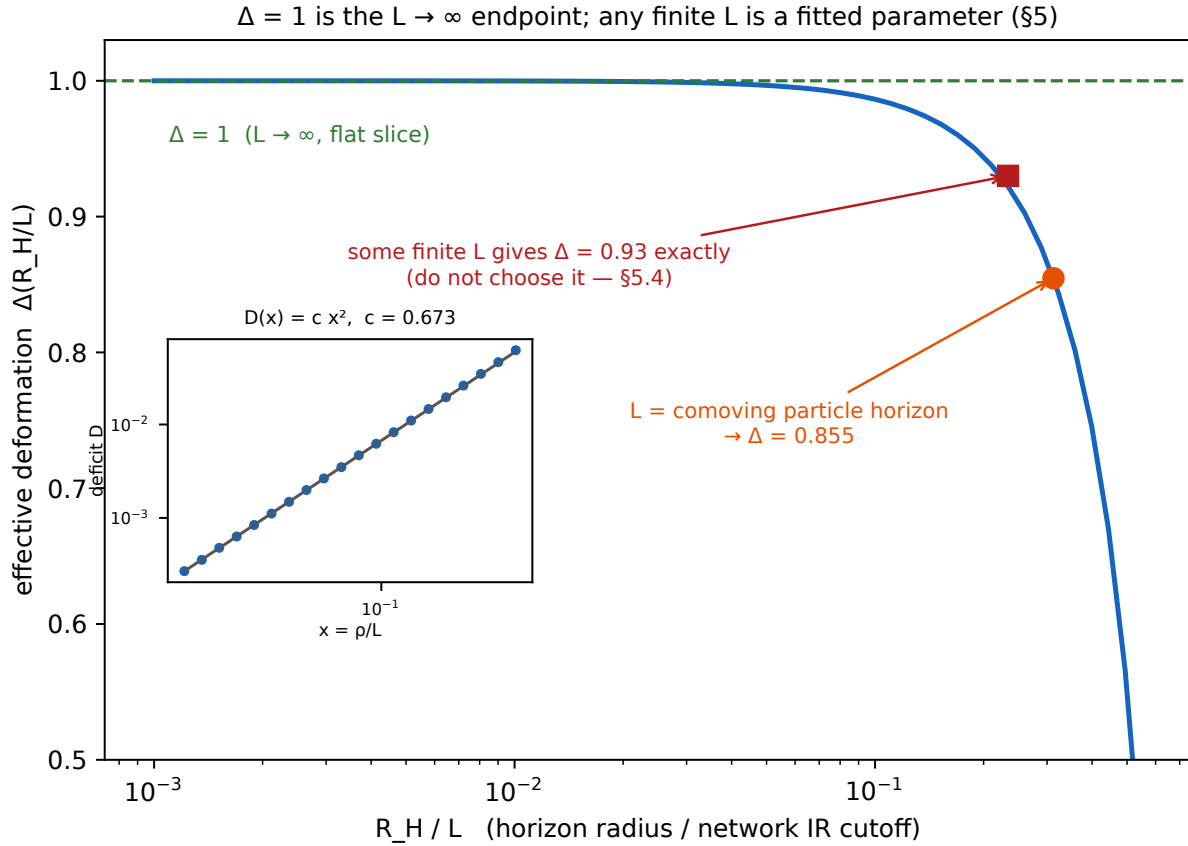


Figure 4. $\Delta = 1$ and the infrared cutoff (§5). The effective deformation $\Delta(R_H/L)$ rises to 1 as the network cutoff $L \rightarrow \infty$. Nothing in the model fixes L , so any finite value is a fitted dark-sector parameter — and choosing one is choosing the answer: L = the comoving particle horizon returns $\Delta = 0.855$, and some L reproduces the foundations paper’s profile value 0.93 exactly (§5.4 cautions against both). Identifying the flat, infinite spatial slice with the network cutoff takes $L \rightarrow \infty$ and $\Delta = 1$ — the endpoint choice of §5.3, which eliminates the parameter without deriving the value. Inset: the exact deficit law $D(x) = c \cdot x^2$ at $\alpha = 1$ (Newton kernel), $c = 0.673$.

theorem of this counting. (An earlier version of this paper presented $\Delta = 1$ as “exact, with zero free parameters,” full stop; that phrasing conflated eliminating a parameter by a cutoff choice with deriving its value, and is corrected throughout.) The point value agrees with the foundations paper: an earlier $\Delta \in [0.98, 1]$ window there, whose derivation rested on a logarithmic Edwards–Wilkinson roughening of the horizon surface, collapses to the point $\Delta = 1$ once that roughening is removed — as it is both there and in §2.5 here. This paper now grounds that point value in the network capacity, as the $L \rightarrow \infty$ endpoint. There was never a window that is both derived and free.

5.4. A warning we place in the text deliberately

Setting L equal to the comoving particle horizon ($\approx 3.2 R_H$) returns $\Delta = 0.855$. The profile-likelihood interval of the foundations paper’s §7 is $\Delta = 0.93$ [0.83, 1.02]. There exists a choice of L returning 0.93 exactly.

We do not make it, and we recommend that no one does. A prediction is over-determined only when every quantity entering it is fixed independently of the datum it is compared against. Here the only principle available fixes $L = \infty$ and hence $\Delta = 1$; any other value of L is selected by the answer. The same discipline applies to a second numerical coincidence in this programme: the volume well of the foundations paper’s free energy sits at $m^* \approx 0.93$, and m is an amplitude while Δ is an exponent. Two coincidences at the same value, neither carrying information.

5.5. Pre-registration

Table 5: Pre-registered predictions for the Barrow deformation Δ and the conditions under which the framework is falsified.

statement	value	status
driven cosmic horizon	$\Delta = 1$ exactly	pre-registered point; the $L \rightarrow \infty$ endpoint reading (a postulate, §5.3), no free parameter once the endpoint is adopted
equilibrium horizon (black hole)	$\Delta = 0$	the foundations paper’s §5(iii), unchanged
intermediate Δ	not produced by any gravitational ($1/r, \alpha < d$) or local coupling	falsifies the framework (§2.4)
$\Delta > 1$	impossible	falsifies the framework (§2.2)

DESI DR3 + Euclid reach a forecast $\sigma(\Delta) \approx 0.09$ (Fisher, hence optimistic): $\Delta = 1$ separates from $\Delta = 0$ at a nominal $\sim 11\sigma$ and from any intermediate value at $\gtrsim 4\sigma$.

6. From capacity to entropy: the state factor and the gate

Everything in §2 concerns the *capacity* $C(\Omega)$ — the bond-cut upper bound on the entanglement entropy of a region. Turning a capacity into the horizon entropy S that

enters the cosmology (the cosmology paper's $\rho_{\text{DE}} = T_{\text{AH}} s_{\text{grav}}$) takes one further ingredient, and it is worth isolating exactly what it is and what it is not.

6.1. Attainment is the state question

The bond cut $C(\Omega) = \sum_{i \in \Omega, j \notin \Omega} w_{ij}$ is the Ryu–Takayanagi/minimum-cut *bound*: $S(\Omega) \leq C(\Omega)$, with equality only for a state that saturates it. So the whole of §2 — the theorem $\Delta \leq 1$, the {area, volume} dichotomy, the $\alpha = d$ crossover — is *state-independent*; but the identification $S = C$, which fixes the *value* of the entropy rather than its scaling, is not. It holds when the horizon degrees of freedom are maximally entangled (thermalised, Page-saturated), so that every bond across the cut carries its full $\ln \chi$. That is precisely the *state* half of the postulate, argued independently in the foundations paper (route A: Sachdev–Ye–Kitaev (SYK) / maximal scrambling) and untouched here. We are explicit that this is the one thing §2 does not remove: the network fixes the *count* — which degrees of freedom a region's entropy may draw on — while the maximally-scrambled state fixes that they are all in fact used. §2 makes the count a theorem; the value of the entropy still rests on the state.

6.2. The entropy density and the occupancy gate

On the volume branch ($\alpha < d$) with the saturating state, $S(\rho) = C(\rho) = \kappa n V(\rho)$, so the entropy density is the constant

$$s_0 = \kappa n = \kappa / \ell_P^3 \quad (6.1)$$

of eq. (2.7) — the cosmology paper's s_{grav} , legs per Planck cell. A horizon does not, however, carry its full volume capacity at every epoch: the capacity is *populated* as structure drives it. Writing $f_{\text{sat}} \in [0, 1]$ for the activated fraction,

$$S = f_{\text{sat}} s_0 V, \quad (6.2)$$

the driven limit $f_{\text{sat}} \rightarrow 1$ is the volume law ($\Delta = 1$) and the undriven limit $f_{\text{sat}} \rightarrow 0$ is the area baseline ($\Delta = 0$) — the state-dependent count of the foundations paper's §7.1, now with s_0 supplied by the network rather than posited. The gate is not a new function: minimal absorption kinetics — each deposit activates a fraction of the *remaining* inactive capacity, $df = \gamma(1 - f) d(D^2)$ — integrate to $f_{\text{sat}} = 1 - e^{-\gamma D^2}$, exactly the cosmology paper's gate (the foundations paper's §7.1a).

6.3. Occupancy is not the order parameter (a correction to Fig. 3 of the foundations paper)

It is tempting to identify f_{sat} with the order parameter m of the area $\leftrightarrow$ volume free energy $F(m)$ of the foundations paper's §5. They are different variables, and their dynamics part company at the endpoint. The order parameter is a *free-energy* coordinate: it relaxes by gradient flow, $\partial_\tau m = -\Gamma F'(m)$, a *branch-selecting* dynamics fixed by the shape of F and

driven by the pitchfork at the horizon’s birth. The occupancy is a *cumulative* coordinate: it obeys the gate kinetics $\partial_x f = \gamma(1 - f)$ (§6.2), whose rate vanishes as $f \rightarrow 1$ — a monotone filling that can never overshoot and carries no barrier. A variable whose own equation of motion switches off at the endpoint it is approaching cannot be the order parameter of a double-well potential. The division of labour is therefore clean: the *branch* — which well the horizon sits in — is fixed at birth by m (the foundations paper’s §5 iii); f_{sat} is the subsequent *occupancy* of the already-selected volume capacity, not a second branch selector. Insofar as the presentation of Fig. 3 of the foundations paper invites its landscape to be read against f_{sat} , the landscape variable is the order parameter m ; f_{sat} is the distinct occupancy coordinate of (6.2).

6.4. The loop to the cosmology paper

Equations (6.1)–(6.2) close the chain from the count to the observable with no fitted dark-sector parameter. The entropy density $s_0 = \kappa n$ is fixed by unitarity (legs per Planck cell); the exponent $\Delta = 1$ by §2 (the volume branch) and §5 (the flat slice, $L \rightarrow \infty$ — the endpoint choice); the gate f_{sat} by the derived kinetics of §6.2. These are exactly the two dark-sector inputs the cosmology paper’s horizon-fluid ansatz $\rho_{\text{DE}} = T_{\text{AH}} s_{\text{grav}}$ requires — the volume-law entropy density and its occupancy — so the cosmology paper’s background is reproduced with every dark-sector quantity now grounded in the network count. What is *not* discharged, and we say so plainly, is the state assumption of §6.1: §2 makes the ceiling and the dichotomy theorems, but the value of the horizon entropy is attained only for the maximally-scrambled state carried over from the foundations paper. That single assumption is the honest residue of the capacity→entropy step.

7. The residue

Table 6: The assumption ledger: status of each component of the volume-law input before this paper (the foundations paper’s §8) and after the reduction.

item	status before (the foundations paper’s §8)	status now
state (maximal entanglement)	first-principles (route A / SYK)	unchanged
form ($S \propto V$)	reduces to thermalisation	unchanged
scale ($\rho_{\text{DE}} \sim \rho_{\text{crit}}$)	CKN bound + flatness	unchanged

item	status before (the foundations paper's §8)	status now
count (bulk vs boundary)	open — the residue	reduced (§2, §5): binary $\{0, 1\}$ given the network model; $\Delta = 1$ is the $L \rightarrow \infty$ endpoint choice, still a postulate
geometric ceiling $\Delta \leq 1$	postulate in effective form (cosmology paper, §8.1); reduced to bound-only in the foundations paper, §7.1	theorem (2.3)–(2.4)
$\Delta = d_H - 2$ (fractal horizon)	assumed (cosmology paper, §8.1); withdrawn in the foundations paper	withdrawn (§2.5)
roughening class (EW vs KPZ)	conditional prediction (foundations paper, pre-fold)	no such classification exists
transverse diffusivity coefficient	scoped open coefficient (foundations paper, §5 iv)	not defined (§2.5); the ϵ -lag layer it fed is retracted in the foundations paper (§6)
—	—	<i>new</i> : finite leg budget κ (unitarity)
—	—	<i>new</i> : $w(r)$ decays slower than r^{-d}
—	—	<i>new</i> : the horizon dof are a network
—	—	<i>new</i> : the infinite-cutoff identification $L \rightarrow \infty$ (§5.3)

Four premises remain, and they are weaker than the one they replace. The leg budget is unitarity. The bond-weight condition is an inequality, not a functional form, and gravity satisfies it with two decades of margin in the exponent. The infrared-cutoff identification is a stated endpoint choice — the residue of the original postulate, relocated and made explicit rather than eliminated (§5.3). The state premise is untouched and independently argued. What has gone is the bulk-versus-boundary count *as a free binary* — the item both previous papers name as irreducible is now a binary forced to $\{0, 1\}$ by the network's range dichotomy — together with the fractal reading of Δ and the roughening layer built on it.

We claim a reduction, not a proof. The bond cut is a capacity, an upper bound on entanglement entropy; its attainment is the state question, and the identification of the horizon degrees of freedom with a network of the assumed connectivity is a model of

the horizon, not a derivation of one. What the reduction buys is that the remaining conditionality now sits in one place, is stated as an inequality, and is decided — for the one horizon we have — by a measurement already scheduled.

8. Discussion

Status: this is a complexity/connectivity layer, not a proven entropy derivation (governing note). Read the construction below as a model of coarse-grained horizon *connectivity*, decoupled from any claim to have derived a gravitational entropy or a stress tensor. Three points govern. (i) The volume-scaling quantity is a structural complexity C_{str} , not the generalized entropy S_{gen} : the bond-cut capacity bounds how many degrees of freedom a region’s entropy *draws on*, which need not equal $A/4G + S_{\text{out}}$. (ii) A horizon-tracking CKN cutoff gives an *area-like* $\Delta = 0$ (density $\propto H^2$), not the volume-like $\Delta = 1$; the volume count is the *non-default* identification, allowed by long-rangedness ($\alpha < d$) but not forced, and the one concrete de Sitter model (double-scaled SYK) currently reads as area — so $\Delta = 1$ remains the empirical bet of the cosmology paper, tested by DESI DR3 + Euclid, not a theorem of this counting. (iii) The bridge to cosmology is an *empirical ansatz*, not a derivation:

$$C_{\text{str}} \longrightarrow q_R = \sigma_R^2 \longrightarrow u(q_R),$$

i.e. the complexity is identified with the model-owned filtered matter variance $q_R = \sigma_R^2$ ($R = 8 \, h^{-1} \text{Mpc}$) that gates the FPAB-SEDE background — and its ensemble and renormalization are *not* specified here. We do not infer the cosmological amplitude μ , the filter scale R , the normalization q_* or the braiding fraction b from node counting; those are fixed by the finite cubic-KGB action and the data (the cosmology paper), and all cosmological comparisons below should be read against that action’s predictions, not the earlier smooth-fluid solution.

The de Sitter holography question, addressed constructively. The residue lives in de Sitter holography [16]. The foundations paper reduced the count to its sharpest form: in the crossed-product (Type II₁) description of the de Sitter static patch, the *value* of the maximal horizon entropy is a trace calibration — a free normalisation the algebra does not fix — so “the eternal-patch ceiling is the area” is an Einstein-saddle calibration, not an algebraic output, and the FRW-patch calibration is left open as “the count asked algebraically.” This paper answers it constructively, without solving the operator-algebraic problem: the trace is fixed by the network — κ legs per Planck cell — and the count that trace delivers is *volume*, because the horizon’s connectivity is long-range ($\alpha < d$). We do not claim to have computed $\dim \mathcal{H}(\text{static patch})$ non-perturbatively; we claim that the quantity the II₁ calibration stands in for — how many degrees of freedom a horizon region’s entropy draws on — is a bond count, and that bond count is the enclosed volume for any gravitational horizon. Consistently with the foundations

paper, the algebra *type* is not the discriminator (an accelerating attractor is II_1 -compatible either way); the calibration is, and the calibration is the leg budget.

What is not settled. Two things, stated plainly. First, the $\alpha = d$ crossover of §2.3 is physics-rigorous but not yet a theorem: the dominance argument on the outward-fraction integral (2.5) is correct term by term, but the interchange of the $\rho \rightarrow \infty$ and continuum limits is not justified, and a mathematically-minded reader will want it bounded — this is the one place the *derivation*, as opposed to the numerics, is incomplete. Second, and more fundamental, the identification $S = C$ is the state question (§6.1): the network gives the count, the maximally-scrambled state gives that the count is attained. That assumption is route A of the foundations paper, and it is the honest residue of the whole programme — now a single question, and one shared with area-law black holes (a black hole carries the same maximally-entangled horizon state; what differs is that it is undriven, so its capacity is the boundary count, §6.2). The open content of SEDE is thereby relocated, in its entirety, from the *count* to the *state* — and the state half is settled, independently of SEDE, by the scrambling / Page-curve physics the foundations paper cites.

Falsifiability. Because the count is discrete for a gravitational horizon (§2.4), the prediction is a *point*, $\Delta = 1$, not a fitted exponent: an intermediate Δ is not a universality class to be selected but a refutation, and DESI DR3 + Euclid forecast $\sigma(\Delta) \approx 0.09$ (Fisher) — separating volume from area at a nominal $\sim 11\sigma$ and from any intermediate value at $\gtrsim 4\sigma$. The companion black-hole channel ($\Delta = 0$ for equilibrium horizons, §6.2) makes the framework falsifiable on a *second*, independent horizon: an isolated volume-law black hole would mean the deformation is universal, refuting the state-dependent, driven-versus-equilibrium count. Resting the count on a network makes SEDE more falsifiable, not less.

9. Conclusion

We set out to reduce the one input the foundations paper could not — whether the cosmic horizon’s entropy counts its enclosed bulk or its boundary. Modelling the horizon degrees of freedom as a network with a finite leg budget and long-range bonds, we found the *ceiling* is a theorem and the count binary: the bond-cut capacity bounds $\Delta \leq 1$ identically; the capacity is the enclosed volume for every coupling of range $\alpha < d$ and the boundary area for $\alpha > d + 1$, with gravity’s $1/r$ reach ($\alpha = 1$) deep on the volume side. The value $\Delta = 1$ is not a theorem of this counting: it is the $L \rightarrow \infty$ endpoint of the one free scale — the infrared cutoff, which nothing internal fixes — reached by identifying the flat, infinite spatial slice with the network cutoff, a stated choice under which the pre-registered point carries no free parameter. The fractal reading of Δ is unnecessary (the ceiling is a finite-dimension bound) and, we showed, unavailable (no minimum-cut surface exists for a long-range network, hence no Hausdorff dimension and no roughening) — which retires, across the companion papers, the roughening layer once built on it. What remains is strictly weaker than the postulate it replaces: finite local dimension, a range inequality gravity satisfies with room to spare, the infinite-cutoff identification, and a maximally-

scrambled state. The last two are the residue — a *state* question and an *endpoint* choice, no longer a free counting — the same questions area-law black holes now answer, and the same measurement, Δ to a forecast $\sigma \approx 0.09$ at DESI DR3 + Euclid, that decides them for the one horizon we have.

Reproducibility

Every number above is produced by the accompanying code from a single entry point, with validation assertions at each stage:

- `src/mincut_membrane.py` — maximum-flow minimum cut on the lattice ball; membrane-thickness and shell-fraction diagnostics (§2.5, tables i–ii).
 - `src/capacity_radial.py` — spherical reduction with the exact angular average; the α -crossover table of §2.3 and the large- ρ $C(\rho)/\rho^2$ diagnostic (marginal-log at $\alpha = d+1$ vs clean area at $\alpha > d+1$), asserted.
 - `src/edge_deficit.py` — Newton-kernel solve at $\alpha = 1$; the deficit law and the convergence of c (§5.1).
 - `src/horizon_constraint.py` — symbolic expansion of the apparent-horizon condition, eq. (2.9); verified against the exact mean curvature of a translated sphere.
 - `src/ams_stability.py` — external-standard check: eq. (2.9) is the Andersson–Mars–Simon MOTS stability operator on the FRW round sphere. Independent (surface-of-revolution) derivation of $-(1/\bar{R})(\nabla_S^2 + 2)$, the published spectrum $\ell(\ell + 1) - 2$ with the $\ell = 1$ translation zero mode, and the AMS matter term $8\pi T_{ab}\ell^a k^b = 3H^2 + \dot{H}$ that isolates the slicing caveat.
 - `run_all.py` — single entry point; runs every stage with validation assertions.
 - `make_figures.py` — regenerates the four figures from the same `src/` computations (Figs 1–4 $\rightarrow$ `output/count_fig{0-3}_*.png`).
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Data availability statement

The code that supports the findings of this article is openly available at <https://github.com/spsingularity/sede-count>, with a tagged release archived at Zenodo, DOI [10.5281/zenodo.21525523](https://doi.org/10.5281/zenodo.21525523). Every quantitative claim is reproduced from a single entry point (`run_all.py`, also wired into the repository-wide `reproduce_all.py` as the `netcount` stage), each stage carrying validation assertions that all pass, and the four figures regenerate from the same computations via `make_figures.py`. No observational data were generated or analysed beyond the standard public inputs cited in the companion papers.

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